

SCIENCE EXHIBITION PROJECT

Smart Classroom Energy Saver

Automatic Occupancy-Based Lighting & Fan Control System

"Turning off is easy. Knowing WHEN — that's smart."

PRESENTED BY

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THE SCALE OF THE PROBLEM

Energy wasted in classrooms, every single day

30%

Electricity wasted in schools

from lights & fans left on in empty rooms

6–8 hrs

Daily school hours

manual switching is unreliable throughout

₹50,000+

Potential annual savings

per school, from automated control

40%

Cut in electricity bills possible with automated occupancy-based control of lights and fans

0

Human intervention needed once installed — sensors handle switching automatically

100%

Classrooms covered when scaled across a school with low-cost sensor units

Figures are estimated for a typical mid-size school based on common classroom usage patterns.

PROBLEM STATEMENTS

01 & 02 & 03

PROBLEM 1

Manual Switches Invite Human Error

- Lights & fans are left ON long after a classroom is empty
- Students and teachers simply forget to switch off
- No corrective mechanism exists between classes or after school hours
- Wastage compounds across dozens of rooms every single day

PROBLEM 2

No Real-Time Usage Visibility

- Schools have no data on which rooms consume the most power
- Electricity bills arrive monthly — far too late to act on
- No accountability per classroom, floor, or time of day
- Wasteful patterns go unnoticed and unaddressed for years

PROBLEM 3

Fixed Timers Don't Adapt to Reality

- Simple timer switches ignore whether a room is actually occupied
- They also ignore natural daylight — lights stay on in bright rooms
- One-size-fits-all timing wastes energy on quiet, sunny days
- No feedback loop to learn or adjust classroom-by-classroom

Smart Classroom Energy Saver Unit

Why PIR + LDR — Not Just One Sensor

- PIR motion sensor confirms a room is actually occupied by people
- LDR light sensor checks if natural daylight already makes bulbs unnecessary
- Combining both avoids false triggers from sunlight or passing shadows
- Fans and lights are controlled independently for finer energy savings

Adaptive Auto Shut-Off Logic

If no motion is detected for a configurable idle period (e.g. 2–3 minutes), the microcontroller automatically switches off lights and fans via relays — then instantly re-enables them the moment someone re-enters, without any manual switch.

MODULE 1

Motion Sensing

PIR sensor detects human presence in the room

MODULE 2

Light Sensing

LDR sensor measures ambient daylight levels

MODULE 3

Relay Control

Switches lights & fans ON/OFF automatically

HARDWARE ARCHITECTURE

1. NodeMCU ESP8266

Main microcontroller — WiFi enabled, low cost, drives all sensors & relays.

2. PIR Motion Sensor

Detects human presence/movement within the classroom in real time.

3. LDR Light Sensor

Measures ambient daylight to avoid switching on lights unnecessarily.

4. 2-Channel Relay Module

Independently switches classroom lights and fan circuits.

5. 16x2 LCD Display

Shows live room status — Occupied/Empty, Light ON/OFF, Fan ON/OFF.

6. Miscellaneous

Breadboard/PCB, jumper wires, resistors, 5V power supply, buzzer.

OPTIONAL / FUTURE EXPANSION

PZEM-004T Energy Meter

Real-time voltage, current & power (kWh) logging

RTC Module (DS3231)

Schedules based on class periods & school hours

ESP-NOW / MQTT

Multi-room mesh network for whole-school rollout

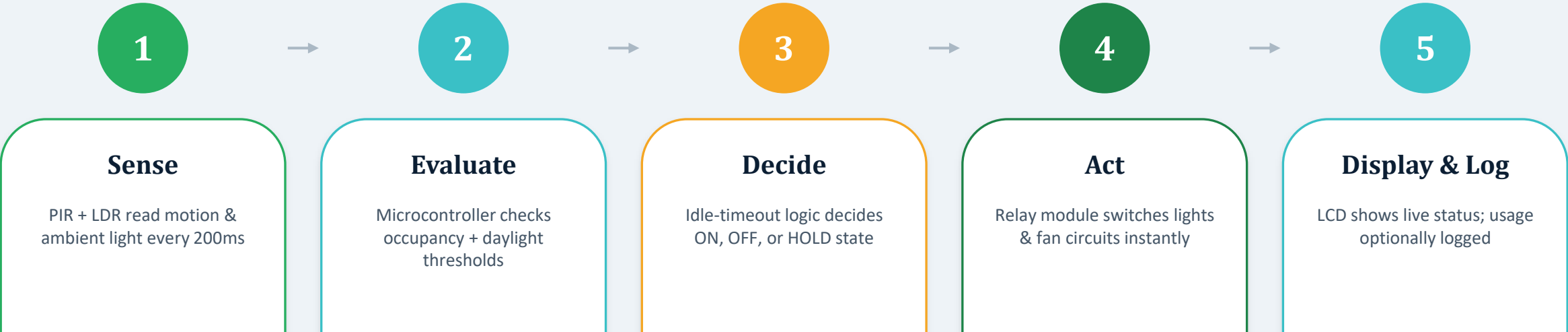
Blynk / Web Dashboard

Remote monitoring & manual override from phone

Buzzer Alert

Reminds staff if a room stays empty with AC/fan running

How Sensing Becomes Automatic Control



CORE CONTROL LOGIC

Occupied `PIR = HIGH → Lights ON only if LDR < threshold`

Daylight Skip `LDR ≥ threshold → Lights stay OFF even if occupied`

Idle Timeout `No motion for T seconds → Fan & Lights OFF`

Re-entry `New motion detected → Instantly restores previous state`

WHERE IT STANDS

vs. existing switching solutions

Solution	Auto Occupancy	Daylight Aware	Live Status	Remote Monitor	Cost (₹)
Manual Switches	No	No	No	No	0
Fixed Timer Switch	No	No	No	No	800+
Basic PIR Light Switch	Yes	No	No	No	600+
Smart Classroom Energy Saver (Ours)	Yes	Yes	Yes	Yes (V2)	~1,200
Commercial BMS System	Yes	Yes	Yes	Yes	1,50,000+

The Smart Classroom Energy Saver delivers full automation at roughly 1/125th the cost of a commercial building management system.

IMPACT & FUTURE SCOPE

30–40% Electricity Savings Achievable per classroom through automated control	₹1,200 Unit Cost Low-cost enough to scale across every classroom	Zero Manual Effort No one needs to remember to switch off again	100% Rooms Covered Scalable to every classroom in the school
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PROJECT ROADMAP

Prototype — Now	PIR + LDR sensing · Relay control · LCD status display
V2 — Next Term	Energy meter logging · RTC scheduling · Web dashboard
Full Rollout — Next Year	Multi-room mesh network · Mobile app · School-wide install

"Every watt saved in a classroom is a lesson in sustainability."